

# Fathi Al Adha Hylmi

## Data Scientist & Machine Learning Engineer

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### PROFESSIONAL SUMMARY

Computer Science graduate (GPA 3.85) with hands-on expertise in machine learning, medical image segmentation, and predictive modeling. Built a Diabetic Retinopathy segmentation pipeline achieving a 0.7755 Macro AUPR and deployed localized health analytics tools that enabled village-level stunting prevention. Proficient in engineering end-to-end ML pipelines using Python, PyTorch, and SQL to solve complex clinical and analytical challenges.

### TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL, C++
- **Machine Learning & AI:** PyTorch, Scikit-learn, Computer Vision (U-Net, DeepLabV3), Predictive Modeling, Genetic Algorithms
- **Data Analysis & BI:** Pandas, NumPy, Power BI, Streamlit, SQL Server, Matplotlib, Seaborn
- **Tools & Frameworks:** Git/GitHub, Linux, CUDA, Jupyter

### MACHINE LEARNING PROJECTS

#### Diabetic Retinopathy Segmentation | Undergraduate Thesis | [\[GitHub\]](#) Python, PyTorch, CUDA, U-Net

- Architected a deep learning pipeline for pixel-level semantic segmentation of 81 high-resolution (1280x1280) Diabetic Retinopathy lesion images utilizing PyTorch and an NVIDIA A100 GPU.
- Mitigated severe class imbalance via a hybrid loss function (BCE, Dice, Focal Loss) and engineered robust data augmentation pipelines (CLAHE, elastic deformations).
- Optimized DeepLabV3 (ResNet-50) to achieve a leading 0.7755 Macro AUPR, while deploying U-Net in parallel to maximize fine-grained microaneurysm localization.

#### Disease Classification System | Independent Project | [\[GitHub\]](#) Python, Scikit-learn, Pandas, SVM

- Engineered an end-to-end machine learning pipeline encompassing exploratory data analysis, missing value imputation, and feature scaling for a multi-disease medical dataset.
- Trained and evaluated classification algorithms, utilizing cross-validation and hyperparameter tuning to systematically optimize predictive performance.
- Achieved a 92% predictive accuracy with an optimized Random Forest model, successfully outperforming the Support Vector Machine (SVM) baseline.

#### Genetic Algorithm Regression Model | Team Project | [\[GitHub\]](#) Python, NumPy, Regression

- Collaborated in a 4-person team to develop a custom Genetic Algorithm optimization framework from scratch to predict complex, non-linear market data applied to the CS:GO economy.
- Designed specialized fitness functions, crossover techniques, and mutation rates to navigate complex search spaces and avoid local optima.
- Demonstrated the algorithm's capability to adapt to dynamic time-series data, providing robust forecasting improvements over baseline linear methods.

### PROFESSIONAL EXPERIENCE

#### Pertamina Hulu Rokan | Front-end Developer Intern DKI Jakarta, Indonesia | Dec 2024 – Mar 2025

Tech Stack: JavaScript, HTML, CSS, .NET

- Automated internal document workflows by engineering a centralized Business Repository, replacing fragmented legacy file-sharing systems and ensuring organizational data integrity.
- Collaborated with graphic designers, senior developers, and management to translate business requirements into scalable technical specifications and responsive UI components.
- Navigated the development lifecycle, integrating stakeholder feedback to optimize the data architecture and improve corporate data accessibility.

#### University Community Service (KKN-PPM UGM) | Technical Lead West Java, Indonesia | Jun 2025 – Aug 2025

Tech Stack: HTML, JavaScript, Web Analytics

- Orchestrated "Digital Empowerment" data initiatives to establish foundational digital infrastructure and data solutions for a local village in West Java.
- Developed a localized health analytics application to process demographic data, cross-referencing government health standards (Permenkes No. 2 Tahun 2020) for stunting prevention.
- Built a scalable digital learning platform and product gallery to facilitate the transition of local enterprises into the digital economy.

### EDUCATION

#### Universitas Gadjah Mada | B.S., Computer Science (GPA: 3.85) Yogyakarta, Indonesia | Aug 2022 - Feb 2026

Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Statistics, Stochastic Processes

#### Nanyang Technological University | Summer Course: Cyber Security (Grade: A+) Singapore | 2024